Dr. Katherine Hawthorne

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PROFESSIONAL SUMMARY

Distinguished Software Architect and Research Fellow with 15+ years of foundational contributions to distributed consensus protocols, high-availability cloud platforms, and large-scale enterprise system transformations. Frequent keynote speaker and author of 6 peer-reviewed papers on distributed systems reliability.

TECHNICAL SKILLS

Languages:	TypeScript, Go, Python, SQL
Frontend:	React, Next.js, Tailwind CSS
Backend:	Node.js, PostgreSQL, Redis
DevOps & Cloud:	Docker, Kubernetes, AWS, Terraform

WORK EXPERIENCE

Staff Platform Engineer @ Apex Cloud Solutions — 2021-03-01 – Present

- Spearheaded architectural re-platforming of legacy monolithic order processing system into 14 distributed Go microservices handling 45,000 req/sec at peak.
- Reduced P99 API latency from 420ms to 28ms by introducing Redis cluster caching layer and gRPC transport protocols.
- Architected automated Kubernetes deployment manifests and Helm charts deployed across 3 AWS multi-region clusters.
- Mentored 8 junior and mid-level engineers, running weekly architecture review sessions and code craftsmanship workshops.

Senior Software Engineer @ Catalyst Financial Technologies — 2018-06-01 – 2021-02-28

- Built and maintained PCI-compliant credit card processing pipeline managing $850M in annual transactional volume.
- Implemented zero-downtime database schema migrations for PostgreSQL tables containing over 250 million records.
- Developed end-to-end audit logging infrastructure using Apache Kafka and Elasticsearch for regulatory compliance.

Full Stack Developer @ Kinetic Web Labs — 2017-06-01 – 2018-05-31

- Engineered customer-facing React SPA dashboard integrated with Express and Node.js REST API endpoints.
- Optimized Webpack bundle sizes by 44% through code-splitting, tree-shaking, and lazy-loading of secondary routes.

Principal Research Scientist @ Global Defense & Aerospace Systems — 2014-01-01 – 2017-05-30

- Directed research team investigating Byzantine fault-tolerant consensus mechanisms for satellite mesh networks.
- Formally verified safety invariants of state machine replication algorithms using TLA+ and formal model checking.
- Awarded 2 US patents on secure multi-hop routing protocols in high-interference electromagnetic environments.

Senior Systems Engineer @ Siemens Industrial Automation — 2010-09-01 – 2013-12-31

- Designed real-time deterministic Ethernet drivers for programmable logic controllers (PLCs).
- Achieved sub-microsecond clock synchronization accuracy adhering to IEEE 1588 Precision Time Protocol.

Member of Technical Staff @ Bell Laboratories 2008-06-01 – 2010-08-31

- Researched optical switching architectures for core telecommunications backbones.
- Co-authored 4 technical whitepapers on dynamic wavelength allocation and network topology optimization.

PROJECTS

Distributed Task Queue Engine Go, Redis, Docker, Prometheus 2023-01-01 – 2023-08-30 github.com/marcusvance/task-queue

- High-throughput async job worker system with Redis persistence, backpressure handling, and exponential backoff retry semantics.

Event Horizon Analytics Node.js, PostgreSQL, WebSockets 2022-03-01 – 2022-11-15 github.com/marcusvance/event-horizon

- Real-time telemetry aggregation engine processing 100k events/sec with WebSockets and TimescaleDB analytics.

Raft-KV Distributed Store Rust, Tokio, gRPC github.com/hawthorne/raft-kv

- Production-ready distributed key-value store implementing Raft consensus with log compaction and dynamic membership changes.

TLA+ Model Repository TLA+, PlusCal github.com/hawthorne/formal-specs

- Curated collection of formal specifications for modern cloud distributed consensus and two-phase commit variants.

EDUCATION

M.S. in Computer Science — **University of Illinois Urbana-Champaign** (GPA: 3.91) 2015-08-15 – 2017-05-20

B.S. in Computer Engineering — **Purdue University** (GPA: 3.78) 2011-08-20 – 2015-05-10

HONORS & CERTIFICATIONS

AWS Certified Solutions Architect – Professional — Amazon Web Services

Certified Kubernetes Administrator (CKA) — Linux Foundation

ACM Distinguished Engineer Award: Awarded for pioneering contributions to distributed system resilience

Best Technical Paper – IEEE CloudCom: Selected among 350 submissions for work on sub-millisecond failover protocols